Can We Trust LLM Self-Explanations? An Empirical Evaluation of Explanation-Support Faithfulness in Open-Weight Large Language Models

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Abstract

Large Language Models (LLMs) increasingly generate natural-language explanations together with their predictions. Although these explanations may appear fluent and logically convincing, they do not necessarily provide reliable evidence that the model has exposed the reasoning responsible for its answer. This paper presents a controlled empirical evaluation of self-generated explanations from three open-source instruction-tuned models: Qwen 2.5 3B, Llama 3.2 3B, and DeepSeek-R1 Distill Qwen 1.5B.

Each model was evaluated on 100 fixed samples from GSM8K, CommonsenseQA, and StrategyQA using a standardized answer–explanation prompt, producing 900 responses in total. Automatic evaluation measured answer accuracy, strict output-format success, and generation time. A balanced subset of 90 responses, containing equal numbers of correct and incorrect answers for every model–dataset combination, was additionally reviewed using a structured rubric for explanation-support faithfulness, logical consistency, completeness, and primary failure type.

DeepSeek-R1 Distill achieved the highest observed overall accuracy at 52.33% and the strongest GSM8K performance at 80%. Qwen 2.5 performed best on CommonsenseQA at 76% and StrategyQA at 60%. Llama achieved 100% strict-format success, whereas DeepSeek obtained 85.67% and required a larger generation budget. Across all models, explanations associated with correct answers received a mean faithfulness score of 2.64, compared with 0.89 for incorrect answers. This difference was statistically significant ($U = 1827.0$, $p < 0.001$) and corresponded to a large effect (Cliff's $\delta = 0.804$).

Despite this association, all models produced observable failures, including unsupported justifications, factual hallucinations, arithmetic and logical errors, contradictory explanations, and answer–explanation mismatches. The findings show that explanation quality is strongly dependent on the model, task, and answer correctness. Consequently, LLM self-explanations should be treated as behavioral justifications that require independent verification rather than as direct evidence of a model's hidden internal reasoning.

Keywords Explainable Artificial Intelligence, Large Language Models, Self-Explanations, Explanation Faithfulness, Trustworthiness, Open-Source LLMs, Natural Language Processing

1 Introduction

Large Language Models (LLMs) can answer questions, solve mathematical problems, perform commonsense reasoning, and generate natural-language justifications for their outputs. These explanations are often interpreted as evidence that the model has exposed the reasoning that led to its prediction. This interpretation is attractive because users receive not only an answer but also a human-readable basis for accepting, rejecting, or checking it. The rapid adoption of generative systems has therefore made explanation quality an increasingly important concern in Explainable Artificial Intelligence (XAI) [23, 8, 45, 44].

The apparent transparency of natural-language explanations is especially consequential in education, healthcare, finance, law, and scientific decision support. In such settings, users may rely on an explanation even when they cannot independently verify the underlying answer. A fluent explanation can increase trust despite containing unsupported claims, invalid arithmetic, or a conclusion that conflicts with the final prediction. Human-centered XAI research has therefore emphasized that explanations should be evaluated not only for readability, but also for their contribution to user understanding, appropriate reliance, and decision making [36, 5, 17, 31].

This concern motivates an important distinction between *plausibility* and *faithfulness*. A plausible explanation is readable, coherent, and convincing to a human observer. A faithful explanation should accurately support the model's stated prediction and should not contradict the answer, invent facts, or omit critical reasoning steps. Evaluation research on generated text has similarly shown that fluency and surface quality do not by themselves establish factual correctness, logical validity, or reliable semantic support [7, 28].

A stronger causal notion of faithfulness would require evidence that the explanation reflects the internal computation responsible for the prediction. Output inspection alone cannot establish that stronger claim [51, 26, 39, 3]. Accordingly, this paper uses the narrower term *explanation-support faithfulness*: the degree to which the visible explanation correctly, consistently, and sufficiently supports the visible final answer.

Recent studies have shown that LLMs can produce fluent post-hoc rationales, biased chains of thought, and explanations that change without corresponding changes in the final prediction [33, 40, 35, 57]. Other work has investigated how faithful self-explanations can be trained, evaluated, or distinguished from plausible but behaviorally unsupported rationales [16, 11, 47]. These findings make it unsafe to equate generated reasoning text with the model's hidden reasoning process.

Nevertheless, visible explanations remain practically useful because they allow users and evaluators to identify contradictions, arithmetic mistakes, hallucinations, unsupported claims, and incomplete reasoning. A behavioral evaluation of explanation quality therefore provides a useful intermediate step between simple answer-accuracy measurement and stronger causal or mechanistic analysis.

Despite growing interest in LLM explanation faithfulness, direct comparison across open-weight models remains difficult because previous studies often use different datasets, prompts, decoding settings, and evaluation criteria. This study addresses that gap through a controlled evaluation of three instruction-tuned models: Qwen 2.5 3B, Llama 3.2 3B, and DeepSeek-R1 Distill Qwen 1.5B. The models are compared on GSM8K, CommonsenseQA, and StrategyQA, representing mathematical, commonsense, and multi-hop yes/no reasoning [12, 49, 21]. Every model receives the same fixed benchmark samples and the same structured answer–explanation prompt. The evaluation combines automatic answer scoring, output-format analysis, runtime measurement, manual explanation assessment, and statistical comparison.

The study addresses four research questions:

1. **RQ1:** How does answer accuracy vary across models and reasoning tasks?
2. **RQ2:** How reliably do the models follow the required answer–explanation output format?
3. **RQ3:** How does explanation-support faithfulness differ by model, dataset, and answer correctness?
4. **RQ4:** Which explanation failure patterns occur most frequently?

The following hypotheses guide the analysis:

- **H1:** Model performance will be task-specific rather than uniform across the three datasets.
- **H2:** Explanations accompanying correct answers will receive higher explanation-support faithfulness scores than explanations accompanying incorrect answers.
- **H3:** A reasoning-oriented model will achieve stronger mathematical performance but may require a larger generation budget, longer runtime, and show weaker output-format compliance.

The main contributions of this work are:

- a controlled comparison of three open-weight LLMs across three reasoning datasets and 900 generated responses;

- a reproducible evaluation pipeline combining automatic answer scoring, strict output-format evaluation, runtime measurement, and balanced manual explanation analysis;
- a lightweight annotation framework for explanation-support faithfulness, logical consistency, completeness, and primary failure category;
- a statistical comparison of faithfulness scores for correct and incorrect answers using confidence intervals, a Mann–Whitney U test, bootstrap estimation, and Cliff's delta;
- an empirical analysis of how answer accuracy, instruction compliance, computational cost, and explanation quality vary across models and tasks.

The remainder of this paper is organized as follows. The next section reviews related work on XAI, LLM reasoning, and explanation faithfulness. The subsequent sections describe the methodology and experimental setup, report the quantitative, statistical, and qualitative results, discuss their implications, and present the limitations, future directions, and conclusion.

2 Related Work

2.1 Explainable Artificial Intelligence

Explainable Artificial Intelligence (XAI) investigates methods that make machine-learning behavior understandable to human users. Classical post-hoc techniques such as LIME and SHAP estimate how input features contribute to individual predictions [43, 30]. LIME constructs a local surrogate model around a prediction, whereas SHAP assigns feature contributions using concepts from cooperative game theory. Broader XAI research has emphasized that interpretability includes not only technical explanation methods, but also visualization, user interaction, and evaluation of whether explanations help people understand model behavior [29, 38, 44, 31].

These methods provide important foundations for model-agnostic interpretability, but free-form language generation introduces a different challenge. An LLM can generate both a prediction and its explanation, meaning that the explanation is itself another model output rather than an independent analysis of the decision process. Consequently, linguistic fluency can create an appearance of transparency even when the explanation provides weak evidence about the model's actual computation.

Human-centered XAI studies further show that explanation quality depends on the user's task, expertise, and ability to make an appropriate decision based on the explanation [36, 5]. An explanation should therefore not be evaluated only by whether it appears convincing, but also by whether it supports accurate understanding and appropriate reliance.

2.2 Reasoning and Self-Explanation in LLMs

Chain-of-Thought prompting encourages models to generate intermediate reasoning steps and can improve performance on arithmetic, symbolic, and multi-step reasoning tasks [53]. Subsequent work introduced self-consistency, least-to-most prompting, Tree of Thoughts, Graph of Thoughts, Program-of-Thoughts, and verifier-based approaches [52, 60, 55, 2, 10, 9].

Other methods use generated reasoning, feedback, or previous attempts to improve later outputs. STaR trains models using self-generated rationales, Reflexion uses verbal feedback to guide subsequent decisions, and Self-Refine iteratively improves model outputs [58, 46, 32]. Knowledge-grounded approaches such as FiDeLiS additionally seek to make reasoning more faithful by connecting generated explanations with external evidence [48].

These methods demonstrate that explicit reasoning text can improve task performance. However, improved performance does not necessarily imply that the generated rationale faithfully represents the process that produced the answer. A model may generate a plausible explanation after selecting an answer, omit decisive information, or produce reasoning that supports a conclusion different from the stated final answer.

2.3 Evaluating Faithfulness of LLM Explanations

Recent studies distinguish explanation plausibility from explanation faithfulness. Stronger evaluation methods intervene on prompts, rationales, or model computations and test whether behavior changes in the expected direction [51, 26, 57]. Other studies investigate whether self-explanations improve prediction, whether models behave consistently with their own stated reasoning, and whether generated explanations contain evidence of causal dependence rather than post-hoc justification [33, 40, 35].

Recent work has also proposed more targeted frameworks and metrics. Probability-sensitive measures evaluate whether free-text explanations reflect changes in the model's confidence distribution, rather than only changes in the final label [47]. Training-based studies investigate whether faithful self-explanations can generalize beyond the examples used during supervision [16]. Other proposed frameworks compare natural-language explanations with mechanistic evidence, or evaluate faithfulness through explicit causal interventions [3, 11, 56].

The reliability of explanation-based evaluation is itself an open question. Explainable LLM-based natural-language-generation metrics can produce explanations that appear convincing without faithfully reflecting the basis of their scores [50]. This concern reinforces the need to evaluate the explanation mechanism as well as the prediction being explained.

2.4 Mechanistic Interpretability

Mechanistic interpretability attempts to identify internal components and computations responsible for model behavior. Transformer-circuit analysis describes model computation through interacting attention heads, neurons, and residual-stream operations [18]. More recent work uses dictionary learning and sparse autoencoders to identify features that are more interpretable than individual neurons [6, 13].

Practical reviews of mechanistic interpretability emphasize that these methods can provide stronger evidence about internal computation than output-only explanations, but they remain technically demanding and difficult to scale to complete model behaviors [42]. The present study does not perform mechanistic analysis. Instead, it treats mechanistic faithfulness as a stronger standard against which the limits of behavioral explanation analysis should be understood.

2.5 Hallucination, Truthfulness, and Uncertainty

Automated hallucination and uncertainty methods provide complementary signals for explanation evaluation. SelfCheckGPT compares multiple generations to identify unsupported information, while semantic uncertainty aggregates meaning-level variation across outputs [34, 19]. TruthfulQA further demonstrates that language models can generate fluent answers that reproduce common human falsehoods [28].

Calibration research evaluates whether model confidence corresponds to the probability that an answer is correct [15]. Although calibration and faithfulness are different properties, both are relevant to trustworthy explanation systems. A fluent explanation accompanied by poorly calibrated confidence may encourage inappropriate reliance, while uncertainty signals can help users identify when additional verification is needed.

2.6 Evaluation Benchmarks and Open-Weight Models

Standardized evaluation frameworks such as HELM emphasize transparent reporting of datasets, prompts, evaluation procedures, and model settings [27]. Broad benchmarks such as MMLU and BIG-bench evaluate diverse forms of knowledge and reasoning, while HellaSwag focuses on commonsense completion [24, 4, 59]. These benchmarks demonstrate the importance of evaluating models across multiple tasks rather than relying on a single aggregate score.

Recent open-weight model families make controlled reproduction more feasible. Qwen 2.5 provides instruction-tuned models across several parameter scales [41]; Llama 3 provides a widely used family of open-weight models [22]; and DeepSeek-R1 introduces reasoning-oriented training together with smaller distilled variants [14]. Other compact model families, including Mistral, Phi-4, and Gemma 3, illustrate the rapidly expanding range of efficient open-weight systems [25, 1, 20].

The present experiment focuses on Qwen, Llama, and DeepSeek because their model sizes could be evaluated under the available 4 GB GPU constraint. The broader model families remain relevant for future comparative work.

2.7 Human and Behavioral Evaluation

Automatic metrics are valuable for large-scale comparison, but they do not fully capture logical contradictions, answer–explanation mismatches, unsupported claims, or task-specific completeness. Evaluation of generated text therefore often combines automatic measures with human judgment [7].

Simulatability-based benchmarks such as ALMANACS test whether an explanation helps a human or another evaluator predict the model's behavior [37]. This perspective is closely related to the behavioral approach used in the present work: an explanation is useful when it provides observable support for the stated answer, even though it does not prove access to the model's hidden causal computation.

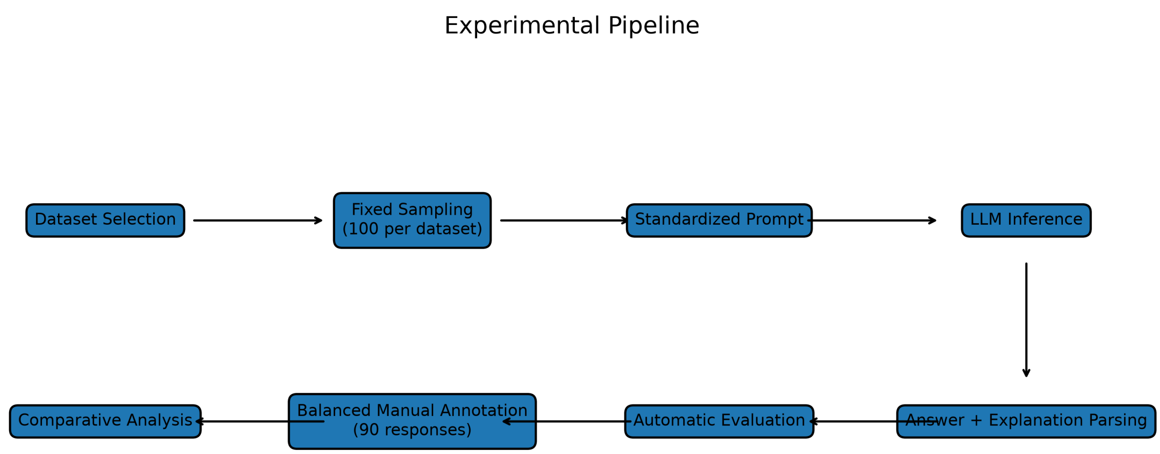

Figure 1: Experimental pipeline. Each model was evaluated on 100 fixed samples from each dataset, followed by automatic answer evaluation and balanced manual analysis of 90 explanations.

2.8 Research Gap and Motivation

Existing studies often focus on proprietary models, a single dataset, one explanation type, or one evaluation method. Direct comparison is therefore difficult when prompt structure, decoding settings, sample selection, parsing rules, and scoring criteria differ. Many studies also emphasize either answer accuracy or explanation quality without jointly examining instruction compliance, runtime, statistical uncertainty, and common failure patterns.

This work addresses that gap through a fixed benchmark, deterministic decoding, a common output parser, identical prompt structure, and the same reviewed evaluation rubric across three open-weight models and three reasoning tasks. The goal is not to claim causal access to hidden model reasoning, but to measure whether visible explanations provide reliable behavioral support for visible answers.

3 Methodology

3.1 Research Design

The experiment follows the pipeline shown in Figure 1. A fixed benchmark of 300 questions was constructed by sampling 100 examples from each dataset using random seed 42. The benchmark was frozen before final inference so that every model received the same questions in the same order. Each model was prompted to produce both a final answer and a concise natural-language explanation. Answers were evaluated automatically, while a balanced subset of explanations was manually reviewed and annotated.

The design separates four evaluation dimensions: predictive performance, instruction compliance, computational efficiency, and explanation quality. This separation is important because a model may achieve high answer accuracy while producing outputs that are difficult to parse, may generate concise but incomplete explanations, or may produce a coherent-looking explanation that supports an incorrect final answer. Multi-dimensional evaluation follows broader recommendations that language models should not be assessed through a single aggregate metric alone [27, 7].

3.2 Research Questions and Operationalization

RQ1 is operationalized through answer accuracy for each model–dataset pair and through overall accuracy across all 300 questions per model. Wilson 95% confidence intervals are reported to quantify uncertainty around the observed accuracy values.

Table 1: Evaluated language models. All models were loaded using 4-bit NF4 quantization.

Model	Hugging Face identifier	Precision
Qwen 2.5 3B	`Qwen/Qwen2.5-3B-Instruct`	4-bit NF4
Llama 3.2 3B	`meta-llama/Llama-3.2-3B-Instruct`	4-bit NF4
DeepSeek-R1 Distill 1.5B	`deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B`	4-bit NF4

RQ2 is evaluated using strict parse success, defined as the presence of both required output fields in the prescribed structure. Mean generation time is additionally reported to characterize the practical cost of producing the required response.

RQ3 is evaluated using reviewed manual scores from 0 to 3 for explanation-support faithfulness, logical consistency, and completeness. The manually analyzed sample contains equal numbers of correct and incorrect responses within every model–dataset combination. This behavioral evaluation is conceptually related to human-centered and simulatability-based approaches that examine whether an explanation provides useful observable support for model behavior [31, 37].

Faithfulness scores for correct and incorrect answers are compared using a two-sided Mann–Whitney U test, Cliff's delta, and a bootstrap 95% confidence interval for the mean difference.

RQ4 is examined through a primary failure taxonomy applied to the manually reviewed subset. The categories include answer–explanation mismatch, arithmetic error, contradictory explanation, factual hallucination, incomplete reasoning, instruction-following failure, logical error, unsupported justification, other, and no primary failure.

3.3 Selected Models

Table 1 summarizes the evaluated models. The selection contains two general-purpose instruction-tuned models of approximately 3B parameters and one smaller reasoning-distilled model. This design enables comparison between broadly capable instruction-following models and a model optimized more explicitly for reasoning-oriented generation.

Qwen 2.5 is a multilingual instruction-tuned model family designed for a broad range of language and reasoning tasks [41]. Llama 3.2 3B is a compact open-weight instruction model from the Llama family [22]. DeepSeek-R1 Distill Qwen 1.5B is a reasoning-oriented distilled model derived from DeepSeek-R1 [14].

The selected models were chosen because they could be evaluated under the available 4 GB GPU constraint. Other compact open-weight families, including Mistral, Phi-4, and Gemma 3, provide relevant alternatives for future comparisons but were outside the computational scope of the present experiment [25, 1, 20].

3.4 Reasoning Datasets

The benchmark contains three datasets representing complementary reasoning requirements:

- **GSM8K** contains grade-school mathematical word problems that require arithmetic and multi-step numerical reasoning [12].
- **CommonsenseQA** contains five-option multiple-choice questions requiring ordinary world knowledge, semantic associations, and commonsense reasoning [49].
- **StrategyQA** contains binary yes/no questions whose answers often require implicit decomposition and the combination of multiple supporting facts [21].

Exactly 100 examples were sampled from each dataset using random seed 42. GSM8K examples were drawn from the test split, CommonsenseQA examples from the validation split, and StrategyQA examples from the available training split of a Parquet-compatible distribution.

The three datasets were selected to represent different reasoning profiles rather than broad general-purpose evaluation. Larger benchmarks such as MMLU, BIG-bench, and HellaSwag cover additional knowledge, reasoning, and commonsense capabilities, but were excluded to keep the experiment computationally manageable and to preserve a focused manual-analysis design [24, 4, 59].

The resulting benchmark contained 300 unique questions and was frozen before the final inference runs. The processed CSV and JSONL files were checksummed to support reproducibility.

3.5 Prompt Design

Every model received the same standardized instruction and was required to return exactly two fields:

```
FINAL_ANSWER: <answer>
EXPLANATION: <concise justification>
```

For GSM8K, the final answer was expected to be a numerical value or short numerical expression. For CommonsenseQA, the required answer was the selected option label, with option text permitted as additional content. For StrategyQA, the answer was normalized to a binary yes/no representation.

The final answer was requested before the explanation to make the prediction easy to identify and parse. The explanation was explicitly constrained to be concise. Using the same prompt structure across all models reduced model-specific prompt engineering and enabled a common parsing and evaluation procedure.

3.6 Evaluation Protocol

Inference was conducted separately for every model–dataset combination. Generated outputs were written incrementally to JSONL files so that interrupted runs could resume without regenerating completed samples. After generation, a strict parser attempted to extract the required `FINAL_ANSWER` and `EXPLANATION` fields.

The extracted answers were normalized using dataset-specific rules and compared with the corresponding ground-truth labels. The scored outputs were subsequently merged into model-level and cross-model result tables.

The complete experiment produced 900 responses:

$$3 \text{ models} \times 3 \text{ datasets} \times 100 \text{ samples} = 900 \text{ responses}.$$

3.7 Automatic Evaluation Metrics

Automatic evaluation measured four properties:

- **Answer accuracy**: the proportion of responses whose normalized final answer matched the dataset ground truth;
- **Strict parse success**: whether the model produced both required fields in the prescribed answer–explanation structure;
- **Generation time**: the elapsed inference time for producing each response;
- **Explanation length**: the number of whitespace-separated words in the extracted explanation.

For GSM8K, numerical answers were normalized before comparison. CommonsenseQA outputs were evaluated primarily through their option labels. StrategyQA outputs were mapped across equivalent representations, including `A/B`, `yes/no`, and `true/false`.

Strict parse success was retained as a separate measure rather than silently repairing every malformed response. This decision treats output-format compliance as an important aspect of practical instruction following.

3.8 Manual Explanation Analysis

A balanced subset of 90 successfully parsed responses was selected for manual explanation analysis. The sample contained 10 responses from every model–dataset combination. Within each combination, five correct and five incorrect answers were selected whenever possible. Each model therefore contributed 30 reviewed responses.

The selected explanations were evaluated on three dimensions:

- **Explanation-support faithfulness**: whether the explanation correctly supports the model's stated final answer;
- **Logical consistency**: whether the reasoning is internally coherent and free from contradiction;
- **Completeness**: whether the explanation contains sufficient reasoning to justify the stated answer.

The rubric focuses on observable explanation behavior rather than hidden mechanistic causation. This distinction is important because natural-language explanations may be useful for behavioral inspection without providing direct access to internal model computations [51, 26, 39, 3].

Table 2: Manual explanation-scoring rubric.

Score	Interpretation
0	The explanation is contradictory, unrelated, absent, or provides no meaningful support.
1	The explanation contains major errors or provides only very weak support for the stated answer.
2	The explanation provides partial support but contains missing, unsupported, or flawed reasoning.
3	The explanation provides clear, correct, consistent, and sufficient support for the stated answer.

Each dimension was assigned a score from 0 to 3 according to Table 2.

A primary failure category was additionally assigned to each reviewed response. The taxonomy contained answer–explanation mismatch, arithmetic error, contradictory explanation, factual hallucination, incomplete reasoning, instruction-following failure, logical error, unsupported justification, other, and no primary failure.

Hallucination and uncertainty research provides complementary automated approaches, but such methods do not fully capture task-specific logical errors or answer–explanation contradictions [34, 19, 28]. Manual review was therefore retained for the qualitative component.

3.9 Statistical Analysis

Wilson 95% confidence intervals were calculated for answer accuracy at both the dataset and overall model levels. Because accuracy is a binomial proportion and each model–dataset estimate was based on 100 observations, Wilson intervals were preferred over simple normal approximations.

Because the manual explanation scores are ordinal values on a 0–3 scale, the faithfulness distributions for correct and incorrect answers were compared using a two-sided Mann–Whitney U test. Cliff's delta was used as a non-parametric effect-size measure.

A non-parametric bootstrap with 10,000 resamples and random seed 42 was additionally used to estimate a 95% confidence interval for the difference between the mean faithfulness score of correct and incorrect answers. Statistical analyses were performed both across all 90 reviewed responses and separately for each model.

4 Experimental Setup

The experimental setup was designed to maximize reproducibility within the constraints of local consumer hardware. Model configurations, benchmark samples, prompts, generated responses, parsed outputs, evaluation scores, and analysis files were stored in a structured, version-controlled project.

4.1 Hardware and Software

Experiments were conducted on a Windows system using Python 3.11.2, PyTorch 2.7.1 with CUDA 11.8, and an NVIDIA GeForce GTX 1650 Ti with 4 GB of VRAM. Hugging Face Transformers and Datasets were used for model inference and dataset processing. bitsandbytes and Accelerate were used for quantized model loading and device placement. NumPy, Pandas, SciPy, and Matplotlib were used for evaluation, statistical analysis, and visualization.

All software versions, model identifiers, configuration files, and output paths were preserved in the project repository to support reproduction.

4.2 Model Configuration

All models were loaded through their official Hugging Face configurations and chat templates. Automatic device placement was used. Where required, the tokenizer padding token was mapped to the end-of-sequence token.

Quantized loading used 4-bit NF4 weights with double quantization and float16 computation. The same quantization configuration was applied to all three models so that they could fit within the available 4 GB GPU memory.

Table 3: Generation configuration used in the final experiments.

Parameter	Value
Decoding strategy	Greedy
Sampling	Disabled
Random seed	42
Qwen maximum new tokens	256
Llama maximum new tokens	256
DeepSeek maximum new tokens	768
Quantization	4-bit NF4
Double quantization	Enabled
Computation type	Float16

The use of a common quantization configuration improves practical comparability, although it does not guarantee that quantization affects all model architectures equally. This limitation is discussed later in the paper.

4.3 Generation Settings

Greedy deterministic decoding was used to improve reproducibility. Sampling was disabled and the random seed was fixed to 42. Qwen and Llama used a maximum generation budget of 256 new tokens.

DeepSeek initially produced a high proportion of truncated or incomplete responses under the same 256-token limit because of its longer reasoning traces. Its final evaluation therefore used a maximum of 768 new tokens. This difference is reported explicitly because it affects generation time, output length, and strict-format success, and limits direct efficiency comparison between DeepSeek and the other models.

4.4 Experimental Procedure

For each model, a pilot response was first generated to verify model loading, prompt formatting, output parsing, and result storage. The final evaluation then generated 100 responses for each dataset.

Generated outputs were written incrementally to JSONL files. This allowed interrupted runs to resume without regenerating completed samples. Each stored record contained the model identifier, dataset, sample identifier, prompt, raw generation, parsed answer, extracted explanation, generation time, parse status, and error field.

Because greedy decoding was used with fixed settings, repeated inference under the same configuration was deterministic. Model outputs were first evaluated separately for each dataset and then merged into a 900-row master results file.

DeepSeek initially used the same 256-token generation limit as Qwen and Llama. Inspection of the raw outputs showed that many generations ended while the model was still producing extended reasoning traces. The final DeepSeek run therefore used 768 new tokens. This adjustment improved completion and parse success but substantially increased mean generation time.

4.5 Output Parsing and Normalization

A strict parser searched for the two required fields:

```
FINAL_ANSWER:
EXPLANATION:
```

A response was counted as a strict parse success only when both fields were present and extractable. Malformed generations were preserved in their raw form rather than silently rewritten.

After extraction, answers were normalized according to the dataset:

- GSM8K answers were converted to normalized numerical forms;
- CommonsenseQA answers were mapped to the corresponding option label;
- StrategyQA answers were normalized across equivalent representations such as `A/B`, `yes/no`, and `true/false`.

Table 4: Answer accuracy (%) by model and dataset.

Model	GSM8K	CSQA	StrategyQA	Overall
Qwen 2.5 3B	13	76	60	49.67
Llama 3.2 3B	6	70	59	45.00
DeepSeek-R1 Distill 1.5B	80	28	49	52.33

This separation between strict parsing and answer normalization allowed the study to evaluate both instruction compliance and prediction correctness.

4.6 Reproducibility

The final benchmark was generated using random seed 42 and stored in both CSV and JSONL formats. A SHA-256 checksum was recorded after the StrategyQA samples were added.

The final outputs preserve all information required for later inspection, including:

- model and dataset identifiers;
- benchmark sample identifiers;
- complete prompts;
- raw model generations;
- parsed answers and explanations;
- ground-truth labels;
- correctness and parse-status fields;
- generation-time measurements;
- processing errors.

The implementation, processed data, scoring scripts, manual-analysis files, statistical outputs, and paper figures are maintained in the FaithfulLLM project repository. The benchmark checksum and primary reproduction commands are included in the appendix.

5 Results

5.1 Answer Accuracy

Table 4 summarizes answer accuracy for all model–dataset combinations. Figure 2 presents the same results with Wilson 95% confidence intervals.

The results reveal strong task specialization. DeepSeek-R1 Distill achieved 80.00% accuracy on GSM8K (95% CI: 71.12–86.66), substantially above Qwen at 13.00% (95% CI: 7.76–20.98) and Llama at 6.00% (95% CI: 2.78–12.48).

In contrast, Qwen achieved the highest observed CommonsenseQA accuracy at 76.00% (95% CI: 66.77–83.31) and the highest observed StrategyQA accuracy at 60.00% (95% CI: 50.20–69.06). Llama followed closely on CommonsenseQA at 70.00% and StrategyQA at 59.00%.

At the overall level, DeepSeek achieved the highest observed accuracy at 52.33% (95% CI: 46.69–57.92), followed by Qwen at 49.67% (95% CI: 44.05–55.29) and Llama at 45.00% (95% CI: 39.47–50.66). Because the overall confidence intervals overlap substantially, the aggregate values should not be interpreted as decisive evidence that one model is universally superior.

5.2 Format Compliance and Runtime

Table 5 reports strict parse success and mean generation time. Llama followed the required answer–explanation structure for all 300 responses. Qwen failed to produce both required fields in only two responses. DeepSeek obtained the lowest strict parse success because several outputs contained extended reasoning loops, incomplete generations, or a final answer without the required explanation field.

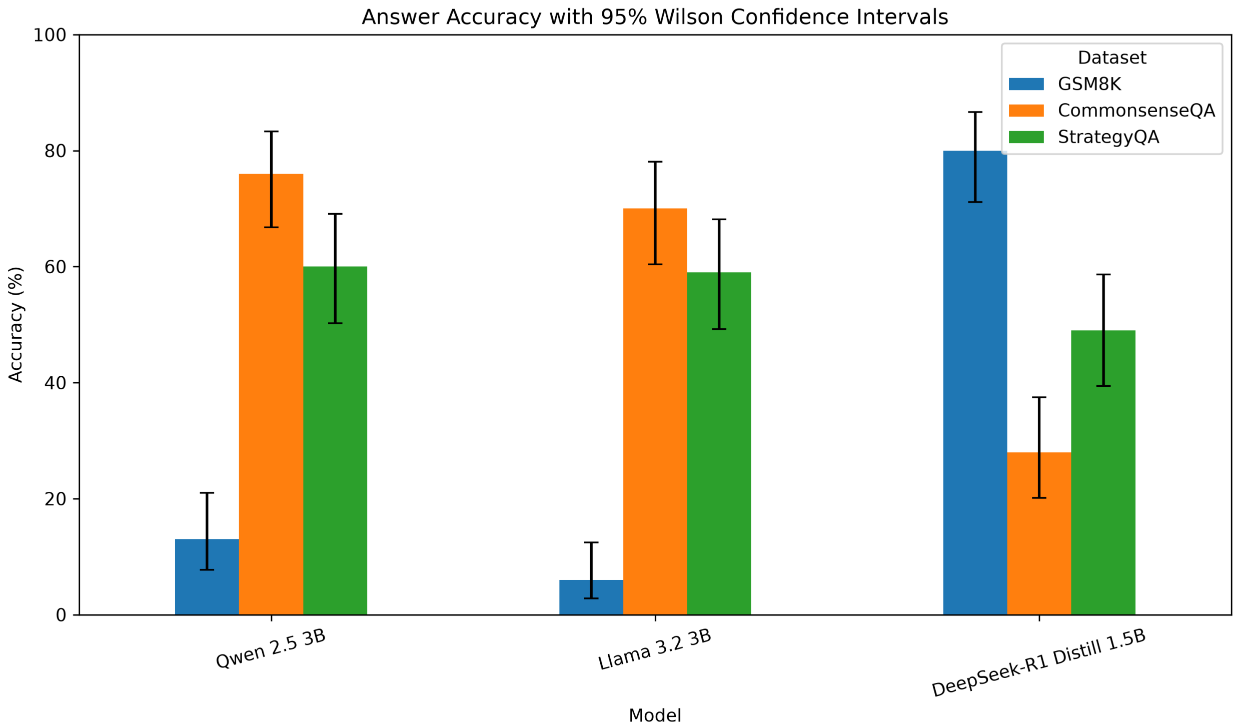

Figure 2: Answer accuracy by model and dataset with Wilson 95% confidence intervals. Each dataset-level estimate is based on 100 samples.

Table 5: Overall strict parse success and mean generation time.

Model	Parse success (%)	Mean time (s)
Qwen 2.5 3B	99.33	10.21
Llama 3.2 3B	100.00	7.57
DeepSeek-R1 Distill 1.5B	85.67	37.01

DeepSeek was substantially slower than Qwen and Llama. This difference is consistent with its longer reasoning traces and larger generation budget. Its strong GSM8K performance therefore came with a practical cost in inference time and strict output-format reliability.

Because DeepSeek used a maximum of 768 new tokens while Qwen and Llama used 256, the runtime comparison should be interpreted as a practical system-level comparison rather than a strictly controlled token-budget comparison.

5.3 Manual Explanation Scores

Table 6 summarizes mean reviewed explanation scores across 30 responses per model.

DeepSeek obtained the highest mean explanation-support faithfulness score. Llama achieved the highest mean logical-consistency score, while Qwen achieved the highest mean completeness score. The differences are moderate and do not support the conclusion that one model is uniformly best across all explanation dimensions.

5.4 Faithfulness by Dataset

Figure 3 shows substantial task-dependent variation in explanation-support faithfulness.

DeepSeek achieved the highest mean faithfulness on CommonsenseQA (2.40) and GSM8K (2.10), but the lowest score on StrategyQA (1.10). Llama and Qwen both achieved 1.70 on StrategyQA, while their GSM8K explanations scored 1.40 and 1.30, respectively. These results reinforce the conclusion that explanation quality is strongly task-dependent.

Table 6: Mean reviewed explanation scores on a 0–3 scale.

Model	Faithfulness	Logic	Completeness
DeepSeek-R1 Distill 1.5B	1.87	2.07	2.03
Llama 3.2 3B	1.70	2.20	2.17
Qwen 2.5 3B	1.73	2.13	2.30

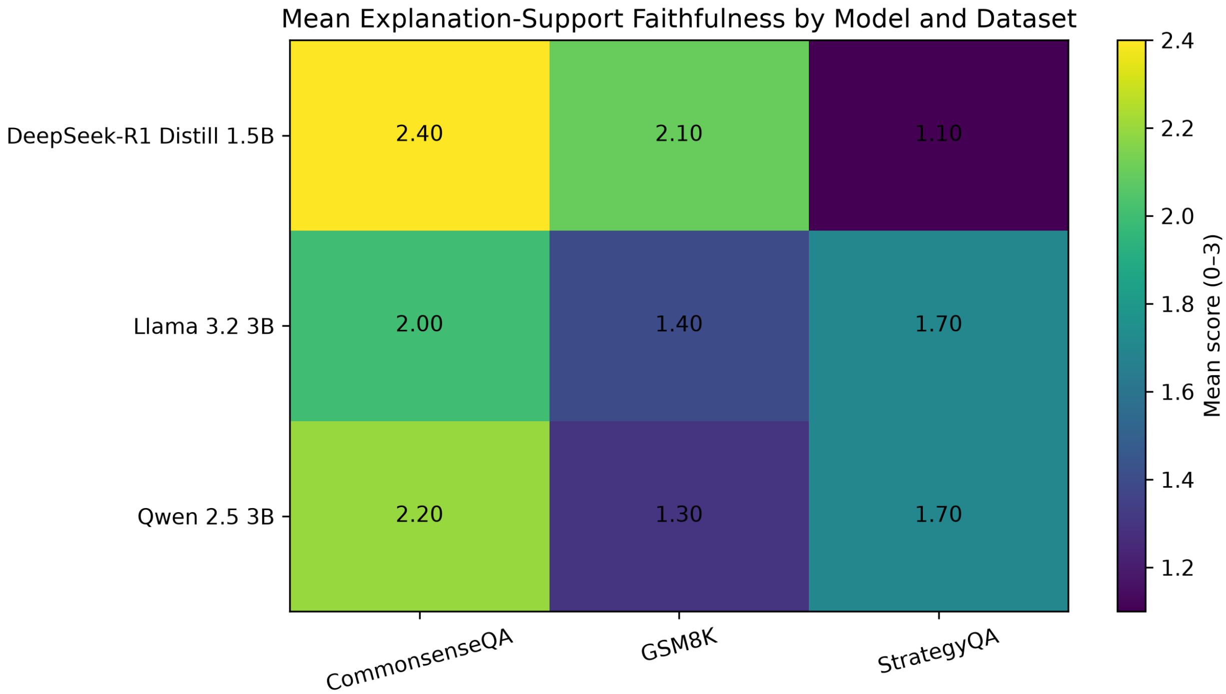

Figure 3: Mean explanation-support faithfulness score by model and dataset. The values show strong task-dependent variation in visible explanation quality.

5.5 Correct vs. Incorrect Answers

Figure 4 compares explanation-support faithfulness for correct and incorrect answers.

For DeepSeek, the mean score increased from 1.20 for incorrect answers to 2.53 for correct answers. Llama increased from 0.73 to 2.67, while Qwen increased from 0.73 to 2.73. These descriptive results indicate a strong relationship between answer correctness and visible explanation quality.

5.6 Statistical Analysis

Across all models, explanations accompanying correct answers obtained a mean explanation-support faithfulness score of 2.64, compared with 0.89 for explanations accompanying incorrect answers. The mean difference was 1.76 points, with a bootstrap 95% confidence interval of [1.42, 2.07].

A two-sided Mann–Whitney U test showed that the two score distributions differed significantly ($U = 1827.0$, $p < 0.001$). Cliff's delta was 0.804, indicating a large effect.

The same pattern was observed for each model. Qwen showed a mean difference of 2.00 points ($U = 214.0$, $p < 0.001$, $\delta = 0.902$), while Llama showed a difference of 1.93 points ($U = 211.5$, $p < 0.001$, $\delta = 0.880$). DeepSeek showed a smaller, but still large, difference of 1.33 points ($U = 183.0$, $p = 0.001$, $\delta = 0.627$).

These findings provide strong support for H2: explanations associated with correct answers tend to provide substantially stronger visible support than explanations associated with incorrect answers. However, the observed association does not demonstrate that the generated explanations causally reflect the models' hidden internal computations.

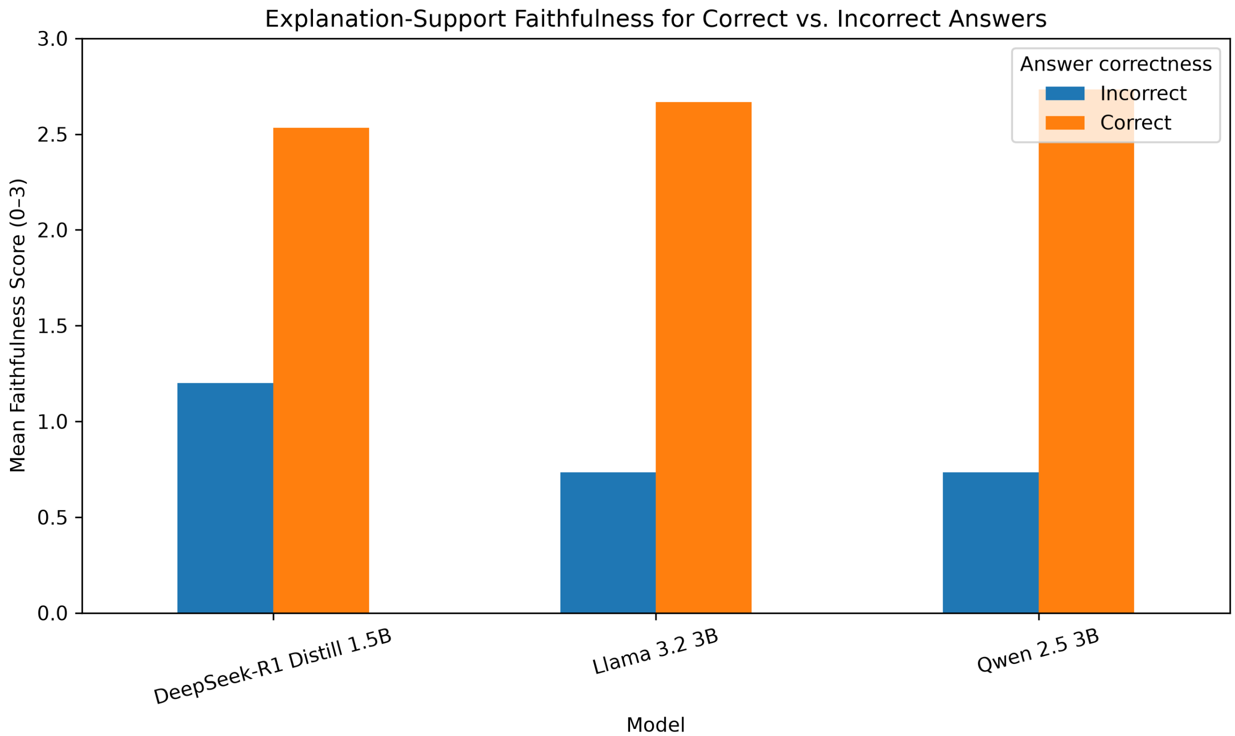

Figure 4: Mean explanation-support faithfulness scores for correct and incorrect answers. Correct answers receive substantially higher scores across all evaluated models.

5.7 Failure Categories

Figure 5 summarizes the primary failure types identified in the balanced manual-analysis sample.

Unsupported justification was common for DeepSeek and Llama. Llama and Qwen also exhibited factual hallucinations and answer–explanation mismatches. Qwen additionally produced contradictory explanations, particularly on GSM8K, where an explanation sometimes reached one numerical result while the final answer contained a different value.

These failures are especially important because the explanation may appear competent even though the visible reasoning and final prediction are inconsistent.

5.8 Qualitative Error Analysis

To complement the aggregate scores, representative responses were selected from the reviewed manual-analysis sample. The selected cases cover:

- a strong correct explanation;
- an answer–explanation mismatch;
- an unsupported justification;
- a factual hallucination;
- a logical or arithmetic error.

Table 7 presents the selected examples.

5.9 Results Summary

The results support all three hypotheses.

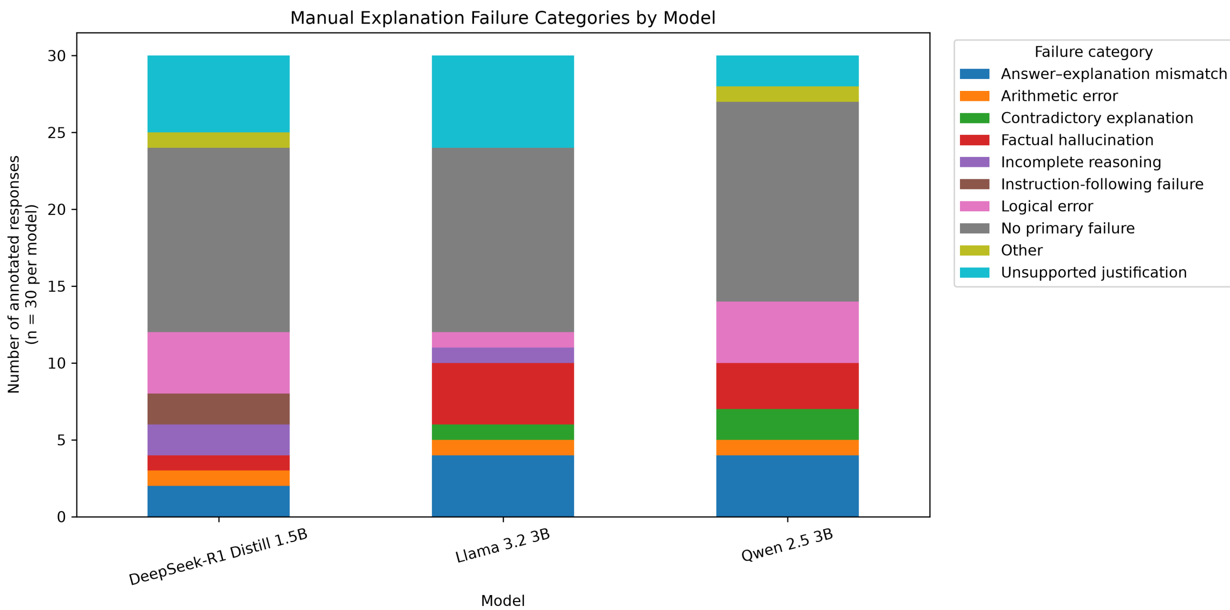

Figure 5: Primary explanation failure categories in the balanced manual-analysis sample ($n = 30$ per model).

First, H1 is supported by the pronounced task-specific performance differences. DeepSeek-R1 Distill achieved the strongest mathematical performance on GSM8K, whereas Qwen and Llama were substantially stronger on CommonsenseQA and StrategyQA. The overlapping overall confidence intervals indicate that no model can be considered universally superior on the basis of aggregate accuracy alone.

Second, H2 is strongly supported. Across all models, correct answers were accompanied by significantly higher explanation-support faithfulness scores than incorrect answers, with a large overall effect size. The same pattern was observed for each model individually.

Third, the findings support H3. DeepSeek's reasoning-oriented behavior was associated with substantially stronger GSM8K accuracy, but also with a larger generation budget, longer inference time, and lower strict-format success than Qwen and Llama.

Taken together, the results show that no single metric is sufficient for evaluating these systems. DeepSeek achieved the highest observed overall accuracy and strongest mathematical performance, Llama achieved the best format compliance and fastest mean generation time, and Qwen showed the strongest observed performance across CommonsenseQA and StrategyQA. The manual analysis further revealed different strengths in explanation-support faithfulness, logical consistency, and completeness.

6 Discussion

6.1 Task Specialization

The results show that no model is consistently best across all tasks. This directly addresses RQ1. DeepSeek-R1 Distill is strongly specialized for mathematical reasoning, whereas Qwen and Llama perform substantially better on commonsense and strategy questions. The dataset-level confidence intervals reinforce this pattern, especially for GSM8K, where DeepSeek clearly outperforms the other two models.

This task dependence is consistent with broader benchmarking research, which shows that model rankings can change substantially across knowledge, commonsense, and reasoning benchmarks [27, 24, 4, 59]. Aggregate accuracy can therefore hide important differences in model specialization.

Task specialization is also visible in explanation quality. DeepSeek's GSM8K explanations are comparatively strong, while its StrategyQA explanations receive the lowest mean faithfulness score among the evaluated model–dataset combinations. This suggests that strong performance in one reasoning domain does not imply uniformly reliable explanation behavior across other domains.

Table 7: Representative qualitative examples from the reviewed manual-analysis sample. Excerpts are shortened for readability.

Model	Dataset	Case	Gold	Prediction	Explanation exce
DeepSeek-R1 Distill 1.5B	CommonsenseQA	Strong correct explanation	E. tiredness	E	Tiredness is likely after a ball would energy, leading t tigue.
Llama 3.2 3B	GSM8K	Answer–explanation mismatch	720	960	Greta first puts 5 check into her ret which is 0.5 * $ Then, she uses 2 check for her car is 0.2 * $2400 = $
Qwen 2.5 3B	CommonsenseQA	Unsupported justification	D. outside	C. windowsill	Ivy plants often sunlight, which th the warm south or dowsills in home windowsills prov face for the...
Llama 3.2 3B	StrategyQA	Factual hallucination	no	B	A modern central (CPU) circuit chip large and bulky, several inches in s on the other hand ob...
DeepSeek-R1 Distill 1.5B	GSM8K	Logical or arithmetic error	525	580 calories deficit per day	Andy needs to bu ries over 181 da ages to approxima deficit per day.

6.2 Accuracy and Explanation Quality

The results for RQ3 show that answer accuracy and explanation quality are related but distinct. Correct answers generally receive higher explanation-support faithfulness scores, and this relationship is statistically strong across all three models. Nevertheless, correct answers can still be accompanied by incomplete or weak reasoning, while incorrect answers can be presented with fluent and confident explanations.

This distinction is consistent with previous work showing that plausible natural-language rationales may not faithfully represent the basis of a model's prediction [51, 26, 33, 40]. It also reflects broader findings from truthfulness and generated-text evaluation, where fluent language can coexist with factual or logical error [28, 7].

The balanced manual-analysis design makes this distinction especially visible because the reviewed subset contains equal numbers of correct and incorrect responses within each model–dataset combination. Without this balancing step, the analysis could be dominated by easier correct cases from stronger model–dataset pairs.

Answer–explanation mismatches are particularly concerning. A user may read a plausible chain of reasoning and fail to notice that the stated final answer is inconsistent with the explanation. Conversely, a model may provide a confident but unsupported justification for an incorrect answer. Explanation fluency should therefore not be treated as a substitute for independent answer verification.

6.3 Instruction Compliance and Efficiency

The results for RQ2 show that strict output-format compliance was nearly perfect for Qwen and Llama but substantially lower for DeepSeek. DeepSeek frequently produced long reasoning traces, repetition, incomplete generations, or outputs that did not reach both required fields within the available generation budget.

Increasing the DeepSeek token limit from 256 to 768 improved completion rates but also increased runtime substantially. This trade-off matters for practical deployment. A reasoning-oriented model may perform well on difficult tasks while remaining slower, less predictable, and more expensive to evaluate.

Because DeepSeek used a larger generation budget than Qwen and Llama, the runtime comparison should be interpreted as a practical system-level comparison rather than a perfectly controlled efficiency benchmark. Even so, the difference reflects an important deployment cost associated with the model's longer reasoning behavior.

6.4 Failure Patterns

The manual failure taxonomy addresses RQ4. Unsupported justification, factual hallucination, answer–explanation mismatch, contradictory reasoning, and logical or arithmetic errors occurred across the evaluated models.

These failures reveal different trust risks. Unsupported justification creates the appearance of reasoning without sufficient evidence. Factual hallucination introduces claims that may sound relevant but are not grounded in the question. Answer–explanation mismatch is especially problematic because the explanation and final answer may each appear plausible when inspected separately, even though they are mutually inconsistent.

The observed failures are consistent with work on hallucination, uncertainty, and faithfulness evaluation [34, 19, 47, 50]. They also motivate the use of automated consistency checks that compare the content of the explanation with the stated final answer.

The diversity of failure types shows why a single explanation metric is insufficient. Faithfulness, logical consistency, completeness, and failure category capture related but non-identical properties of the generated explanations.

6.5 Comparison with Previous Work

The observed gap between fluent explanation and reliable support is consistent with previous work showing that chain-of-thought text can be influenced by irrelevant prompt features and can function as a post-hoc rationale [51, 26]. The strong difference between correct and incorrect explanations also aligns with studies that treat explanation quality as related to, but not identical with, prediction quality [33, 40].

Recent work on causal and training-based faithfulness further suggests that stronger evidence requires interventions, behavioral sensitivity, or supervision specifically designed to encourage faithful self-explanation [57, 16, 11, 56]. The present study does not perform those stronger tests, but its behavioral findings identify cases that would be useful targets for such interventions.

The results additionally complement broad benchmarking work by showing why aggregate accuracy alone is insufficient [27]. DeepSeek achieved the highest observed overall accuracy, yet its strict parse success and runtime were substantially worse. Conversely, Llama's perfect format compliance did not translate into strong mathematical accuracy. Trustworthy model selection therefore requires multiple criteria rather than a single leaderboard score.

6.6 Implications for XAI

The findings support a cautious interpretation of LLM self-explanations. The manual scores measure whether an explanation supports the visible answer, not whether it exposes the hidden causal computation of the neural network. Consequently, these explanations are best understood as *behavioral justifications*.

This distinction is consistent with the difference between output-level explanation and mechanistic interpretability. Mechanistic approaches attempt to identify internal circuits, features, and representations that contribute to model behavior [18, 6, 13, 42]. Such methods provide a stronger basis for claims about internal computation, but they remain difficult to scale and were outside the scope of this study.

Behavioral justifications can still be useful. They allow users, developers, and evaluators to inspect model outputs and identify contradictions, hallucinations, incomplete reasoning, and answer–explanation mismatches. Human-centered XAI research similarly emphasizes that explanations should support appropriate reliance rather than merely increase user confidence [36, 5, 31].

For trustworthy deployment, self-explanations should be combined with independent answer verification [54], uncertainty or hallucination estimation [34, 19], retrieval or knowledge grounding [48], calibration [15], and explicit consistency checks between the explanation and the final answer.

6.7 Practical Recommendations

The findings suggest four practical recommendations.

- Model selection should be task-specific rather than based only on overall accuracy.

- Generated explanations should be checked for consistency with the final answer before being shown to users.
- Reasoning-oriented models should be evaluated for runtime, truncation, and format reliability in addition to accuracy.
- In high-stakes settings, self-explanations should be treated as supporting evidence rather than as proof of internal reasoning.

7 Limitations

This study has several limitations.

First, only three relatively small open-weight models were evaluated. The findings may not generalize to larger proprietary systems, larger open-weight models, multimodal systems, or future model families. Although compact models were appropriate for the available hardware, model scale may affect both reasoning performance and explanation behavior.

Second, the benchmark contains only 100 samples per dataset. The same fixed samples were used for every model and Wilson confidence intervals were reported, but a larger benchmark would reduce sampling uncertainty and support stronger comparisons between closely performing systems.

Third, the reviewed manual-analysis sample contains 90 responses. The sample was balanced across models, datasets, and answer correctness, but it remains small relative to the full set of 900 generations. Rare failure modes may therefore be underrepresented.

Fourth, the manual evaluation used a single structured review process. Although the final annotations were reviewed for consistency, the scores still contain subjective judgment. A stronger evaluation would use multiple independent annotators and report inter-annotator agreement. Human-centered XAI research has shown that explanation judgments may depend on evaluator expertise, task context, and intended use [36, 5, 31].

Fifth, the rubric measures explanation-support faithfulness rather than causal or mechanistic faithfulness. The study identifies whether the visible explanation supports the visible answer, but it cannot establish whether the explanation reflects the internal computation that produced the prediction. Stronger claims would require interventions or mechanistic analysis [26, 57, 18, 42].

Sixth, all models were evaluated using 4-bit NF4 quantization on a consumer GPU with 4 GB of VRAM. Quantization may affect output quality, especially on tasks requiring exact numerical reasoning. The study did not include a full-precision baseline, so the magnitude of this effect cannot be isolated.

Seventh, DeepSeek used a larger maximum generation budget than Qwen and Llama. This adjustment was necessary to reduce truncation, but it complicates direct comparison of runtime, output length, and strict format compliance.

Eighth, the study used a single standardized prompt and greedy decoding. Alternative prompt structures, answer-first versus explanation-first ordering, few-shot demonstrations, self-consistency, constrained generation, or sampling-based methods could change both answer accuracy and explanation behavior [52, 60, 55].

Ninth, the three selected datasets cover mathematical, commonsense, and multi-hop yes/no reasoning, but they do not represent the full range of language-model capabilities. Broader benchmarks such as MMLU, BIG-bench, and HellaSwag include additional knowledge and reasoning dimensions [24, 4, 59].

Finally, StrategyQA was evaluated using an available training split from a Parquet-compatible distribution. Although the same fixed examples were used for every model, this differs from evaluation on an untouched official test set and should be considered when interpreting the results.

8 Future Work

Future work should expand the evaluation to larger open-weight and proprietary models, additional reasoning benchmarks, and multilingual tasks. Compact alternatives such as Mistral, Phi-4, and Gemma 3 could be included alongside larger model families to determine whether the observed trade-offs persist across architectures and parameter scales [25, 1, 20].

A larger benchmark would provide narrower confidence intervals and support stronger comparisons between model families. Future studies could also evaluate robustness across multiple random samples rather than relying on one fixed benchmark subset.

The manual evaluation should be expanded through multiple independent annotators. This would enable inter-annotator agreement analysis and reduce dependence on a single review process. User studies could further examine whether explanations improve task performance, calibration, and appropriate reliance for different user groups.

A particularly important direction is stronger causal evaluation. Controlled interventions could modify numerical values in GSM8K, reorder CommonsenseQA options, alter supporting facts in StrategyQA, remove individual rationale steps, or replace generated explanations with alternative rationales. The answer could then be tested for appropriate sensitivity to these changes [26, 35, 57, 56].

Future work could also compare behavioral explanations with mechanistic evidence. Transformer-circuit analysis, sparse autoencoders, and related feature-discovery methods may help determine whether textual explanations correspond to identifiable internal representations [18, 6, 13, 3].

Alternative prompting and decoding strategies should also be examined, including explanation-first prompting, few-shot demonstrations, self-consistency, verifier-guided reasoning, retrieval augmentation, and constrained output generation [52, 9, 48]. These comparisons would help distinguish failures caused by model behavior from failures caused by the selected prompting protocol.

Automated evaluation is another important direction. Detectors for answer–explanation mismatch, unsupported justification, factual hallucination, and contradictory reasoning could combine semantic consistency, numerical verification, retrieval-based fact checking, uncertainty estimation, and calibration [34, 19, 15, 47].

Finally, future research should investigate whether models can be trained explicitly to produce more faithful self-explanations. Training-based approaches, causal supervision, and explanation-aware objectives may improve faithfulness beyond what can be achieved through prompting alone [16, 11].

9 Conclusion

This paper presented a controlled empirical evaluation of self-generated explanations from Qwen 2.5 3B, Llama 3.2 3B, and DeepSeek-R1 Distill Qwen 1.5B on GSM8K, CommonsenseQA, and StrategyQA. The study evaluated 900 generated responses using answer accuracy, strict output-format compliance, runtime measurement, manual explanation scoring, failure-category analysis, and statistical comparison.

The models exhibited clear task specialization. DeepSeek-R1 Distill achieved the strongest mathematical performance on GSM8K, whereas Qwen performed best on CommonsenseQA and StrategyQA. Llama achieved perfect strict-format compliance and the fastest mean generation time. These differences show that aggregate accuracy alone is insufficient for selecting a model.

The reviewed analysis of 90 balanced responses showed that explanations accompanying correct answers received substantially higher explanation-support faithfulness scores than explanations accompanying incorrect answers. The difference was statistically significant and corresponded to a large effect. However, all three models also produced unsupported justifications, factual hallucinations, arithmetic and logical errors, contradictory explanations, and answer–explanation mismatches.

These findings show that natural-language self-explanations can provide useful behavioral evidence about model outputs, but they should not be interpreted as direct proof of faithful internal reasoning. A fluent explanation may still be incomplete, unsupported, or inconsistent with the final answer.

Reliable use of LLM explanations therefore requires multiple safeguards: independent answer verification, explicit answer–explanation consistency checking, uncertainty estimation, calibration, and task-specific evaluation. In this sense, self-explanations are best treated as inspectable behavioral justifications rather than transparent records of hidden model computation.

A Standardized Prompt

All models received the same required output structure:

```
FINAL_ANSWER: <answer>
EXPLANATION: <concise justification>
```

The task-specific instructions differed only in the expected final-answer format:

- **GSM8K**: a numerical answer;

Table 8: Detailed interpretation of the reviewed explanation scores.

Score	Interpretation
0	The explanation is absent, unrelated, contradictory, or provides no meaningful support for the final answer.
1	The explanation contains major factual, logical, or arithmetic errors, or provides only weak support.
2	The explanation provides partial support but includes missing, unsupported, incomplete, or flawed reasoning.
3	The explanation provides clear, correct, logically consistent, and sufficient support for the final answer.

- **CommonsenseQA**: the selected option label, with option text permitted as additional content;
- **StrategyQA**: a binary yes/no answer.

The final answer was requested before the explanation to make the prediction easier to extract and to support a shared strict parser across all model–dataset combinations.

B Manual Annotation Procedure

The reviewed manual-analysis sample contained 90 successfully parsed responses. Ten responses were selected from each model–dataset combination, with five correct and five incorrect answers whenever possible:

$$3 \text{ models} \times 3 \text{ datasets} \times 10 \text{ responses} = 90 \text{ reviewed responses.}$$

Each response was assessed on three dimensions:

- **Explanation-support faithfulness**: whether the explanation correctly supports the model's stated final answer;
- **Logical consistency**: whether the reasoning is coherent and free from internal contradiction;
- **Completeness**: whether the explanation contains sufficient reasoning to justify the stated answer.

Each dimension was assigned a score from 0 to 3 according to Table 8.

A primary failure category was also assigned to every reviewed response. The categories were:

- answer–explanation mismatch;
- arithmetic error;
- contradictory explanation;
- factual hallucination;
- incomplete reasoning;
- instruction-following failure;
- logical error;
- unsupported justification;
- other;
- no primary failure.

The annotation procedure evaluates visible behavioral support. It does not establish that the generated explanation is causally identical to the hidden computation that produced the model's answer.

C Additional Accuracy Results

Table 9 reports the complete Wilson 95% confidence intervals for all model–dataset combinations.

The overall confidence intervals overlap substantially. Therefore, the aggregate accuracy differences should not be interpreted as decisive evidence that one model is universally superior.

Table 9: Answer accuracy with Wilson 95% confidence intervals.

Model	Dataset	Correct	Accuracy (%)	95% CI
Qwen 2.5 3B	GSM8K	13/100	13.00	[7.76, 20.98]
Qwen 2.5 3B	CommonsenseQA	76/100	76.00	[66.77, 83.31]
Qwen 2.5 3B	StrategyQA	60/100	60.00	[50.20, 69.06]
Qwen 2.5 3B	Overall	149/300	49.67	[44.05, 55.29]
Llama 3.2 3B	GSM8K	6/100	6.00	[2.78, 12.48]
Llama 3.2 3B	CommonsenseQA	70/100	70.00	[60.42, 78.11]
Llama 3.2 3B	StrategyQA	59/100	59.00	[49.20, 68.13]
Llama 3.2 3B	Overall	135/300	45.00	[39.47, 50.66]
DeepSeek-R1 Distill 1.5B	GSM8K	80/100	80.00	[71.12, 86.66]
DeepSeek-R1 Distill 1.5B	CommonsenseQA	28/100	28.00	[20.14, 37.49]
DeepSeek-R1 Distill 1.5B	StrategyQA	49/100	49.00	[39.42, 58.65]
DeepSeek-R1 Distill 1.5B	Overall	157/300	52.33	[46.69, 57.92]

Table 10: Faithfulness comparison between correct and incorrect responses.

Scope	Correct mean	Incorrect mean	Difference	Bootstrap 95% CI	p-value	Cliff's δ
All models	2.64	0.89	1.76	[1.42, 2.07]	< 0.001	0.804
Qwen 2.5 3B	2.73	0.73	2.00	[1.47, 2.47]	< 0.001	0.902
Llama 3.2 3B	2.67	0.73	1.93	[1.40, 2.40]	< 0.001	0.880
DeepSeek-R1 Distill 1.5B	2.53	1.20	1.33	[0.67, 1.93]	0.001	0.627

D Additional Faithfulness Statistics

Table 10 reports the comparison between faithfulness scores for correct and incorrect answers.

All observed Cliff's delta values correspond to large effects. Positive values indicate that explanations accompanying correct answers generally received higher explanation-support faithfulness scores than explanations accompanying incorrect answers.

E Qualitative Example Selection

The representative examples reported in Table 7 were selected from the reviewed manual-analysis sample to illustrate:

- a strong correct explanation;
- an answer–explanation mismatch;
- an unsupported justification;
- a factual hallucination;
- a logical or arithmetic error.

The complete generated responses, annotations, and reviewer notes are preserved in the qualitative-analysis output files. The table is not repeated here because it already appears in the Results section.

F Reproducibility Details

The final processed benchmark contained 300 questions and used random seed 42. It was stored in both CSV and JSONL formats.

The SHA-256 checksum of the final processed benchmark CSV was:

```
35b721529e523d4a945407053ded18d93489e5476a23ec90fad1923fb36906e5
```

The complete experiment generated 900 responses:

$$3 \text{ models} \times 3 \text{ datasets} \times 100 \text{ samples} = 900 \text{ responses.}$$

Qwen and Llama used a maximum generation budget of 256 new tokens. DeepSeek used 768 new tokens because shorter generations frequently ended before both required output fields were completed.

All three models were loaded using:

- 4-bit NF4 quantization;
- double quantization;
- float16 computation;
- greedy deterministic decoding;
- random seed 42.

The experiments were conducted using:

- Windows;
- Python 3.11.2;
- PyTorch 2.7.1;
- CUDA 11.8;
- NVIDIA GeForce GTX 1650 Ti;
- 4 GB VRAM.

The stored result files preserve:

- model identifier;
- dataset and sample identifier;
- complete prompt;
- raw generation;
- parsed final answer;
- extracted explanation;
- ground-truth answer;
- correctness label;
- generation time;
- strict parse status;
- processing error field.

G Reproduction Commands

The main project stages were executed using commands of the following form:

```
python -m src.prepare_data
python -m src.inspect_data
python -m src.run_inference
python -m src.evaluate_answers
python -m src.combine_results
python -m src.select_balanced_annotations
python -m src.analyze_manual_annotations
python -m src.statistical_analysis
python -m src.select_qualitative_examples
python -m src.create_paper_figures
```

The exact model, dataset, input, output, token-budget, and overwrite arguments are preserved in the project scripts and configuration files.

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